**Horizon — runtime pipeline (one request)**
indirect injection = Stage 1 + Stage 2 (Horizon math) · direct injection / jailbreak = rose layer

**Stage 1 — keyed spotlighting (offline, Slides 11–27)**

**Data**
BIPIA clean docs · LLMail-Inject + BIPIA canary attacks

**8 defenses τ, fresh key κ**
none · delimit · ^ · dyn. marker $N^K$ · ROT13
shift 25 · scramble |Σ|! · Base64

**Forging**
$q / |K_\tau|$ (log-space)

**ASR bound**
Clopper–Pearson $\text{Beta}^{-1}(1-\alpha; s+1, n-s)$

**χ²-DRO**
$\bar{p} + \sqrt{(2\rho\ \bar{p}(1-\bar{p}))}$

**Adaptive attacker**
$\text{argmax}_{\delta \in \Delta_B} E_K[A] \to \bar{p}^{adp}$

**Combined risk**
$\bar{R}_\tau(\rho) = \min\{1, q/|K| + \max(\text{DRO}, \text{adaptive})\}$

**Choice rule**
$\tau^* = \text{argmin}\ \bar{R}$ s.t. $U \geq U_{none} - \eta$

**Sion game, level 1**
$V_1$ SOCP on paired $A[x, \tau] \to \pi^*$

**Stage 1 outputs**
$\pi^* \cdot |K| = \min |K_\tau| \cdot \bar{R}_Q = \Sigma\ \pi^*\ \bar{R}_\tau$

**User prompt u**
trusted channel

**Input guard**
DeBERTa injection classifier (ProtectAI)

**Privileged LLM**
sees ONLY u → writes a program

**External data**
emails · web · tool outputs (untrusted)

**CaMeL interpreter**
tracks data + control-flow dependencies

π*, |K|

untrusted x

**Quarantined LLM × m**
each run: own τ ~ π*, own fresh κ
no tools · typed JSON schema

$\bar{R}Q$

**Forging**
$\Sigma\ \pi^* \cdot q / |K_\tau|$

**QIF leakage**
$\leq \Sigma \log_2|S|$ bits · cap per config

**Majority vote**
> m/2 agree, else refuse · score s(x) = agreement

**Hoeffding**
$\exp(-2m(\frac{1}{2} - \bar{R}_Q)^2)$ · vs empirical

**Typed value + capability**
v = (data, type, cap) · cap ∈ {SYSTEM ⊑ USER ⊑ UNTRUSTED}

**Policy check**
YAML rule table per tool + role rules

**Noninterference**
Z3 proof at startup: UNTRUSTED ⇒ ¬ALLOW

**ALLOW**
policy met

**CONFIRM → ask user**
only if s(x) ≥ λ̂, else DENY

**Conformal risk control**
$\hat{\lambda}$: $(n/(n+1))\hat{R} + 1/(n+1) \leq \alpha_{user}$

**DENY**
violation

approved

**Tool call**
AgentDojo environment

per-attack G, C, F

per-attack G, C, F

per-attack G, C, F

**Output guard**
canary token check + Qwen3Guard

**Horizon risk + Sion game**
loss $\ell = G + a \cdot C + h \cdot F$ · Sion LP → $\pi^*$ (Horizon), w*
$V\ (\text{Horizon}) = \min\{1, \Sigma\pi\ q/|K| + \max(E[L] + \sqrt{(2\rho\ \text{Var}\ L)}, \Sigma\pi\ \ell\ adp)\}$

**Answer**

**Legend**

**LLM / spotlighting**

**Math tool**

**Code / data**

**Direct / jailbreak layer**
not in the Horizon math